

12. (2 points)

An insurer is using the following information without adjustment to build a stochastic loss reserve model for its property exposures in a coastal area:

Accident Year	Number of Hurricanes	Number of All Other Peril Events	Losses from Hurricanes (\$000)	Losses from All Other Perils (\$000)
2012	0	4,750	0	54,625
2013	1	5,150	250,000	60,255
2014	0	5,000	0	60,000
2015	0	5,300	0	65,190
2016	0	4,900	0	61,250
Total	1	25,100	250,000	301,320

- The data is valued as of June 30, 2017.
- The underlying exposures for the book of business are not expected to change.
- The resulting model produced an R^2 value greater than 99%.
- The insurer will perform a retrospective test of the model using ten years of data valued as of June 30, 2027.
- Assume no data errors.

a. (1 point)

Identify and briefly describe two sources of internal systemic risk applicable to this model.

b. (1 point)

Draw a histogram assuming no hurricane losses in the next ten years and briefly explain your selection of the shape.

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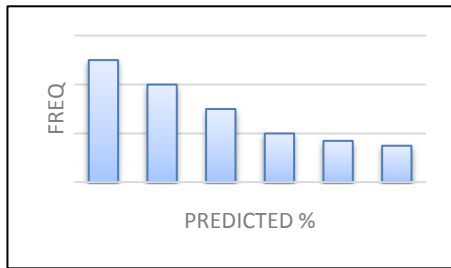
EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

SPRING 2018 EXAM 7, QUESTION 12	
TOTAL POINT VALUE: 2	LEARNING OBJECTIVE(S): A5
SAMPLE ANSWERS	
Part a: 1 point	
<u>Sample 1</u> • Specification error: The risk that the model cannot fully capture the complexities of the insurance process. • Parameter selection error: The risk that the selected parameters & their trends do not match the true parameters & their trends. <u>Sample 2</u> • Specification error: Catastrophes are complex to model and it will be difficult to accurately capture the intricacies in an internal model. • Parameter estimation error: We're looking at a tiny amount of data for hurricanes; 5 years is not enough time to get an accurate frequency view. Severity will also be difficult to estimate as there is only 1 event. <u>Sample 3</u> • Specification error: the error resulting from the insurance process being too complex for any model to fully capture. This is applicable because hurricane & other CAT exposures are very volatile & complex. • Parameter selection error: the error from being unable to adequately measure predictors of claim cost outcomes & trends in these predictors. Insurer has very limited information to fit parameters from (only 1 hurricane & only going to test 10 years of data). <u>Sample 4</u> • Specification risk: only one model is used in generating results. You could improve it by using multiple models to capture strengths of various models. • Parameter error: The selected parameters could be based on historical data that has changed in recent years. Could cause an understatement in future losses. <u>Sample 5</u> • Specification error: did we use multiple models to get to our answer or just one? Best practice would be to consider several models. • Parameter selection: Are our predictors generated close to the best predictors? Have we identified the best predictors? Data seems thin so I'm doubtful.	

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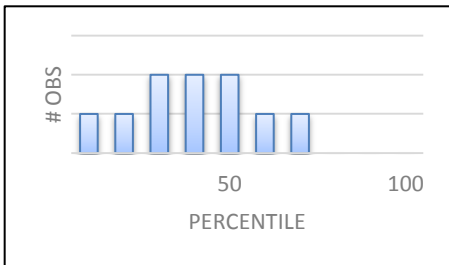
Part b: 1 point

Sample 1



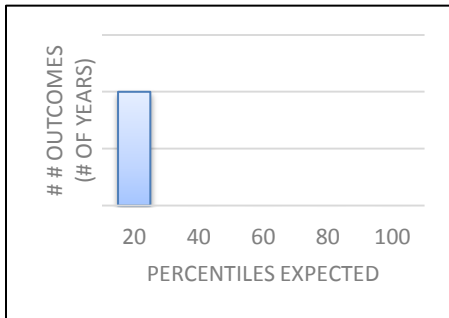
Model overestimates losses, as it expects cats in a couple years at least. With no cats, losses come in lower than expected, so our model is predicting too high (biased high).

Sample 2



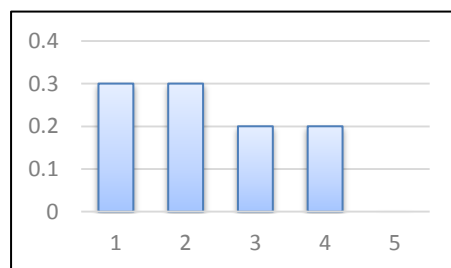
The model results would appear biased high. Most actual outcomes would be in low percentiles due to absence of cat losses.

Sample 3



Compared to a period where hurricane losses made up almost 80% of losses, non-hurricane years will always be much lower than the model predictions.

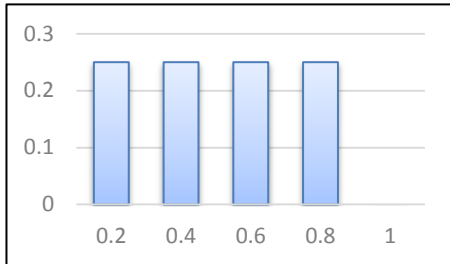
Sample 4



I expect most results would fall in the lower expected percentiles because there were no hurricanes.

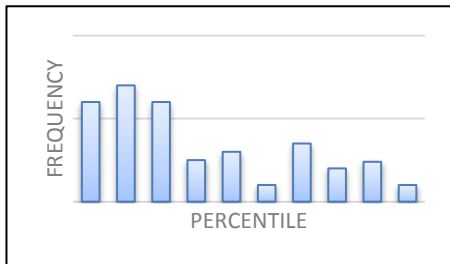
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Sample 5



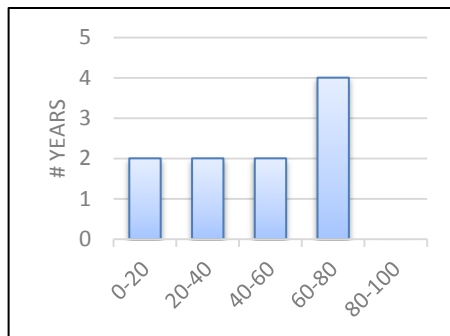
Based on historical data, hurricane happens 1 in 5 years. Since no hurricane in the next ten years, 0 in (80%-100%) and 0.25 in (0,20%), (20%,40%),...

Sample 6



Model will be biased high (due to no hurricane losses) and is likely overfit. There will be more volatility due to overfitting. (Shape at higher percentiles due to overfitting).

Sample 7



Hurricanes are rare, and my guess is that including the hurricane losses without adjustment in only 5 years of data has caused the model to be overfit and biased high. Since hurricanes occurred in 1 out of 5 years in the data, all years in the test period will be low 80th percentile of distribution. The other losses will probably be too low because there is a positive severity trend that wasn't adjusted for in the data, so below 80th percentile, model is biased low, so 60-80 will be higher.

EXAMINER'S REPORT

Candidates were expected to describe the systemic risks present in the construction of a stochastic reserve model and evaluate their potential impact on the model prediction by means of a histogram showing actual future outcomes versus the percentiles implied by the stochastic reserve model.

Part a

Candidates were expected to identify and briefly describe the two applicable internal systemic risks, Specification Error & Parameter Selection Error.

Common mistakes included:

- Other risks/errors named (e.g. model risk, estimation error, projection error, data quality risk, event risk, process risk).
- Vague or incomplete descriptions of Specification Error (e.g. risk of not selecting the correct/right model).

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- Risk of using one model.
- Mentioning Data Error even though the question specifically stated to assume no data errors.

Part b

Candidates were expected to draw and explain a hypothetical histogram from the results of a retrospective test conducted on this stochastic reserve model using the next 10 years of data. Given that the model was fit with a high correlation coefficient on an experience period including a hurricane event which generated a high volume of losses, and the next 10 years gave rise to no hurricane events, that would imply a high bias, or possibly heavy tail, manifesting itself in a histogram that has more mass in the lower percentiles anticipated by the model. There was wide latitude in the types of histograms and associated shapes accepted, as long as the histogram considered the data and implications of actual vs. expected emergence with an appropriate explanation.

Common mistakes included:

- Providing a chart of the estimated losses ordered by accident year, rather than a histogram with expected loss percentiles or buckets.
- Providing a chart that is not clearly a histogram.
- Providing histograms that show expected losses, but do not consider the retrospective test of the model's prediction.
- Not recognizing the bias that the hurricane loss may cause in the model predictions of future non-hurricane loss years.